

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	264852	49.245289
1	[Git, Change, Create]	99583	18.515977
2	[Git, Change, Delete]	61316	11.400798
3	[Git, Change, Rename]	24820	4.614910
4	[File, Git]	16609	3.088196
5	[Git, Commit]	16390	3.047477
6	[Java, ByteCode, Parameter]	9457	1.758388
7	[Java, ByteCode, Member, Method]	9072	1.686803
8	[Java, ByteCode, Bound]	6545	1.216945
9	[Java, Value, ByteCode, Annotation]	5562	1.034171
10	[Java, ByteCode, Bound, ParameterizedType]	5029	0.935068
11	[Xml, Element]	2512	0.467069
12	[Java, ByteCode, Member, Field]	2238	0.416123
13	[Xml, Text]	1692	0.314602
14	[Java, ByteCode, Member, Constructor, Method]	1591	0.295823
15	[Java, ByteCode, Bound, WildcardType]	1079	0.200624
16	[Java, ByteCode, Bound, TypeVariable]	868	0.161392
17	[Java, ByteCode, Member, Method, Lambda]	839	0.156000
18	[Type, File, Java, ByteCode, JavaType]	698	0.129783
19	[Type, File, Java, ByteCode, Class]	675	0.125506
20	[Git, Change, Copy]	546	0.101521
21	[Json, Key]	523	0.097244
22	[Value, Json, Scalar]	513	0.095385
23	[Java, ByteCode, GenericDeclaration, Member, M...	481	0.089435
24	[Type, File, Java, ByteCode, ResolvedDuplicate...	477	0.088691
25	[Java, Value, ByteCode, Primitive]	364	0.067680
26	[Type, File, Java, ByteCode, ExternalType]	330	0.061359
27	[Author, Git, Person]	307	0.057082
28	[Committer, Git, Person]	248	0.046112
29	[Maven, Dependency]	205	0.038117

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

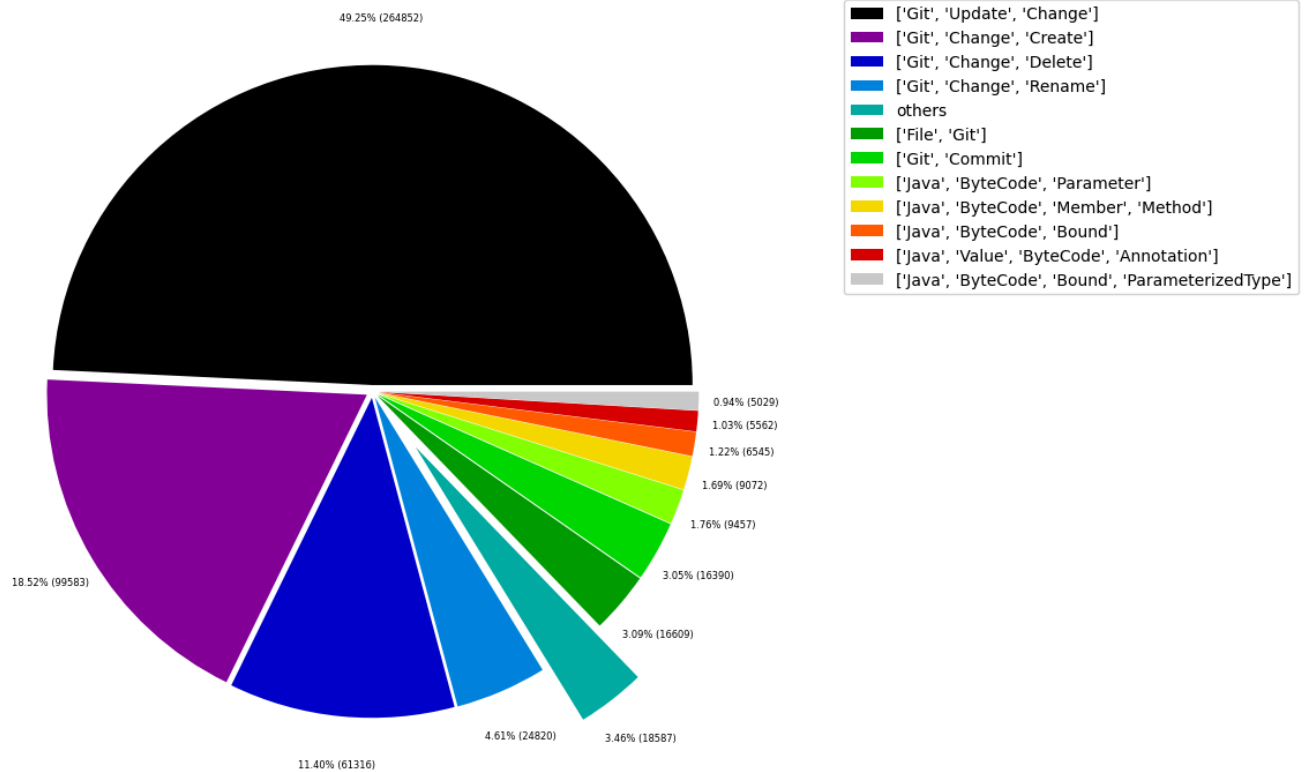


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000186
1	[Git, Branch]	1	0.000186
2	[Repository, File, Git]	1	0.000186
3	[File, Json]	2	0.000372
4	[Type, File, Java, ByteCode, GenericDeclaratio...	4	0.000744
5	[File]	4	0.000744
6	[Maven, Exclusion]	6	0.001116
7	[Value, Array, Json]	8	0.001487
8	[Java, ByteCode, GenericDeclaration, Member, C...	9	0.001673
9	[Type, File, Java, ByteCode, Throwable, Extern...	10	0.001859
10	[File, Maven, Xml, Pom, Document]	11	0.002045
11	[File, Java, Manifest]	11	0.002045
12	[Java, ManifestSection]	11	0.002045
13	[Type, File, Java, ByteCode, Void]	11	0.002045
14	[Maven, Scm]	11	0.002045
15	[File, Artifact, Jar, Archive, Zip, Java]	11	0.002045
16	[Type, File, Java, ByteCode, Throwable, Resolv...	11	0.002045
17	[Maven, ExecutionGoal]	13	0.002417
18	[File, Java, ServiceLoader]	13	0.002417
19	[File, Java, Properties]	13	0.002417
20	[Maven, PluginExecution]	13	0.002417
21	[Type, File, Java, ByteCode, Enum]	17	0.003161
22	[jQAssistant, Rule, Concept]	19	0.003533
23	[Maven, Configuration]	20	0.003719
24	[Maven, Plugin]	20	0.003719
25	[Xml, Attribute]	22	0.004091
26	[Type, File, Java, ByteCode, PrimitiveType]	39	0.007251
27	[Type, File, Java, ByteCode, Class, Throwable]	42	0.007809
28	[Type, File, Java, ByteCode, Annotation]	42	0.007809
29	[Xml, Namespace]	44	0.008181

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

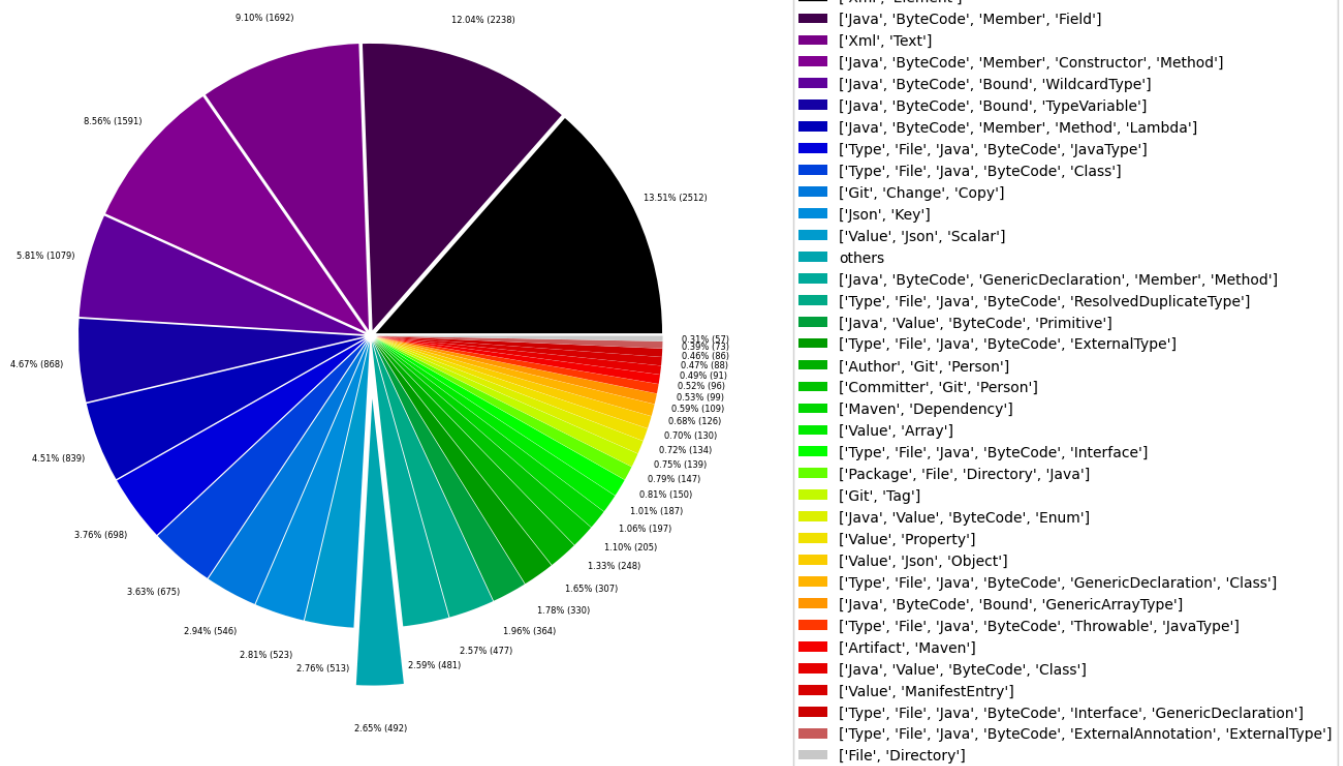


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

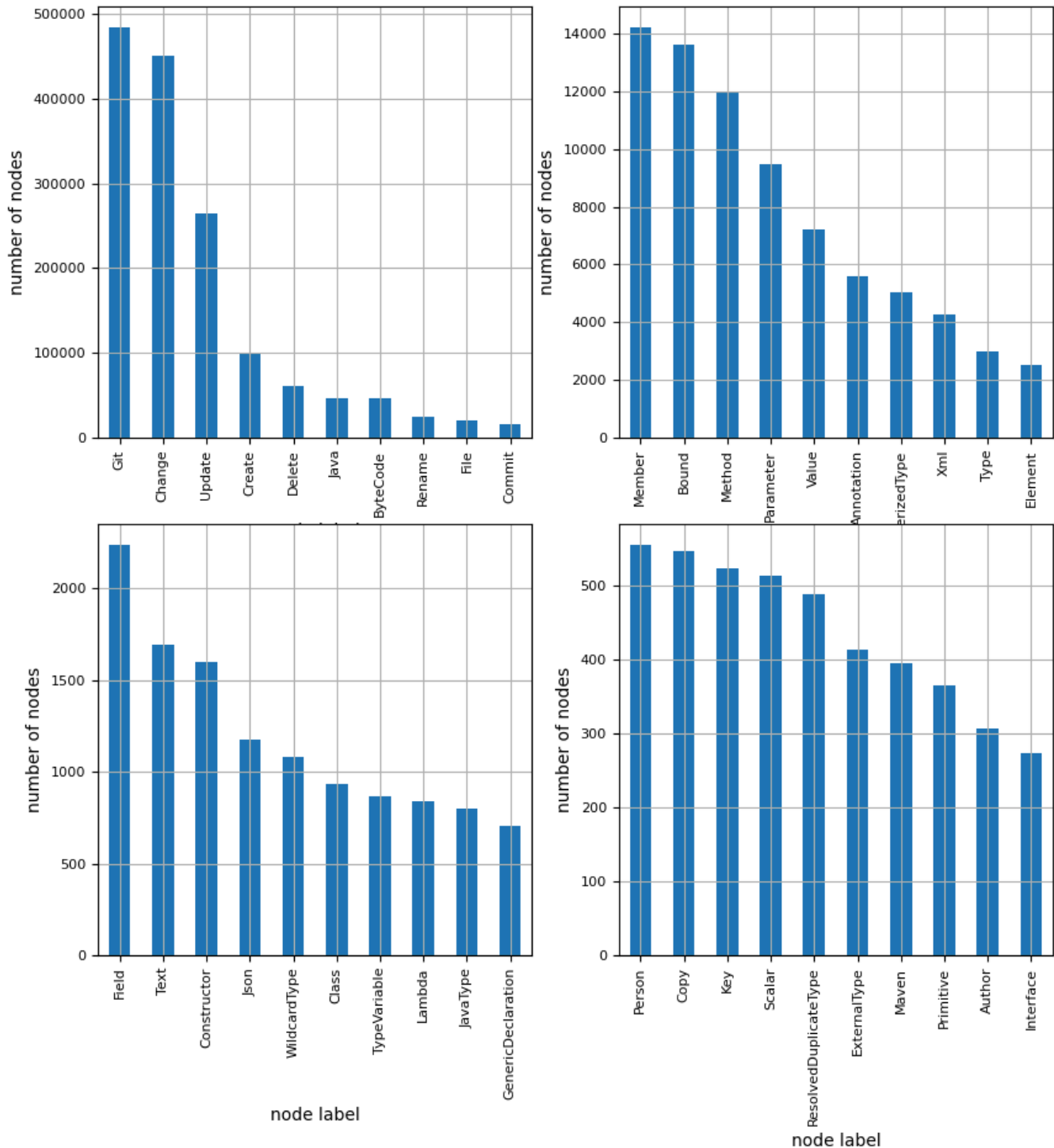
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	484820	90.145067
1	Change	451117	83.878495
2	Update	264852	49.245289
3	Create	99583	18.515977
4	Delete	61316	11.400798
5	Java	46661	8.675919
6	ByteCode	46452	8.637058
7	Rename	24820	4.614910
8	File	19861	3.692857
9	Commit	16390	3.047477
10	Member	14230	2.645857
11	Bound	13630	2.534296
12	Method	11992	2.229734
13	Parameter	9457	1.758388
14	Value	7226	1.343567
15	Annotation	5604	1.041980
16	ParameterizedType	5029	0.935068
17	Xml	4281	0.795988
18	Type	2979	0.553901
19	Element	2512	0.467069
20	Field	2238	0.416123
21	Text	1692	0.314602
22	Constructor	1600	0.297496
23	Json	1176	0.218660
24	WildcardType	1079	0.200624
25	Class	934	0.173663
26	TypeVariable	868	0.161392
27	Lambda	839	0.156000
28	JavaType	797	0.148190
29	GenericDeclaration	706	0.131270
30	Person	555	0.103194
31	Copy	546	0.101521
32	Key	523	0.097244
33	Scalar	513	0.095385
34	ResolvedDuplicateType	488	0.090736
35	ExternalType	413	0.076791
36	Maven	395	0.073444
37	Primitive	364	0.067680
38	Author	307	0.057082
39	Interface	273	0.050760

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1668569

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	451117	27.036161
1	MODIFIES	451117	27.036161
2	UPDATES	264852	15.873003
3	CREATES	124949	7.488393
4	DELETES	86136	5.162268
5	COMMITTED	32780	1.964558
6	RENAMES	24820	1.487502
7	INVOKES	22756	1.363803
8	HAS_PARENT	19906	1.192998
9	OF_TYPE	19011	1.139360
10	DEPENDS_ON	18178	1.089437
11	HAS_FILE	16609	0.995404
12	HAS_COMMIT	16390	0.982279
13	DECLARES	14584	0.874042
14	OF_RAW_TYPE	13531	0.810934
15	HAS	10124	0.606747
16	HAS_NEW_NAME	9599	0.575283
17	RETURNS	8301	0.497492
18	READS	6113	0.366362
19	HAS_ACTUAL_TYPE_ARGUMENT	5936	0.355754
20	ANNOTATED_BY	5561	0.333280
21	CHANGED_TOGETHER_WITH	4683	0.280660
22	OF_GENERIC_TYPE	4539	0.272030
23	SIMILAR	3496	0.209521
24	CONTAINS	3309	0.198314
25	RESOLVES_TO	2698	0.161695
26	RETURNS_GENERIC	2679	0.160557
27	HAS_FIRST_CHILD	2512	0.150548
28	HAS_LAST_CHILD	2512	0.150548
29	HAS_ELEMENT	2490	0.149230

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

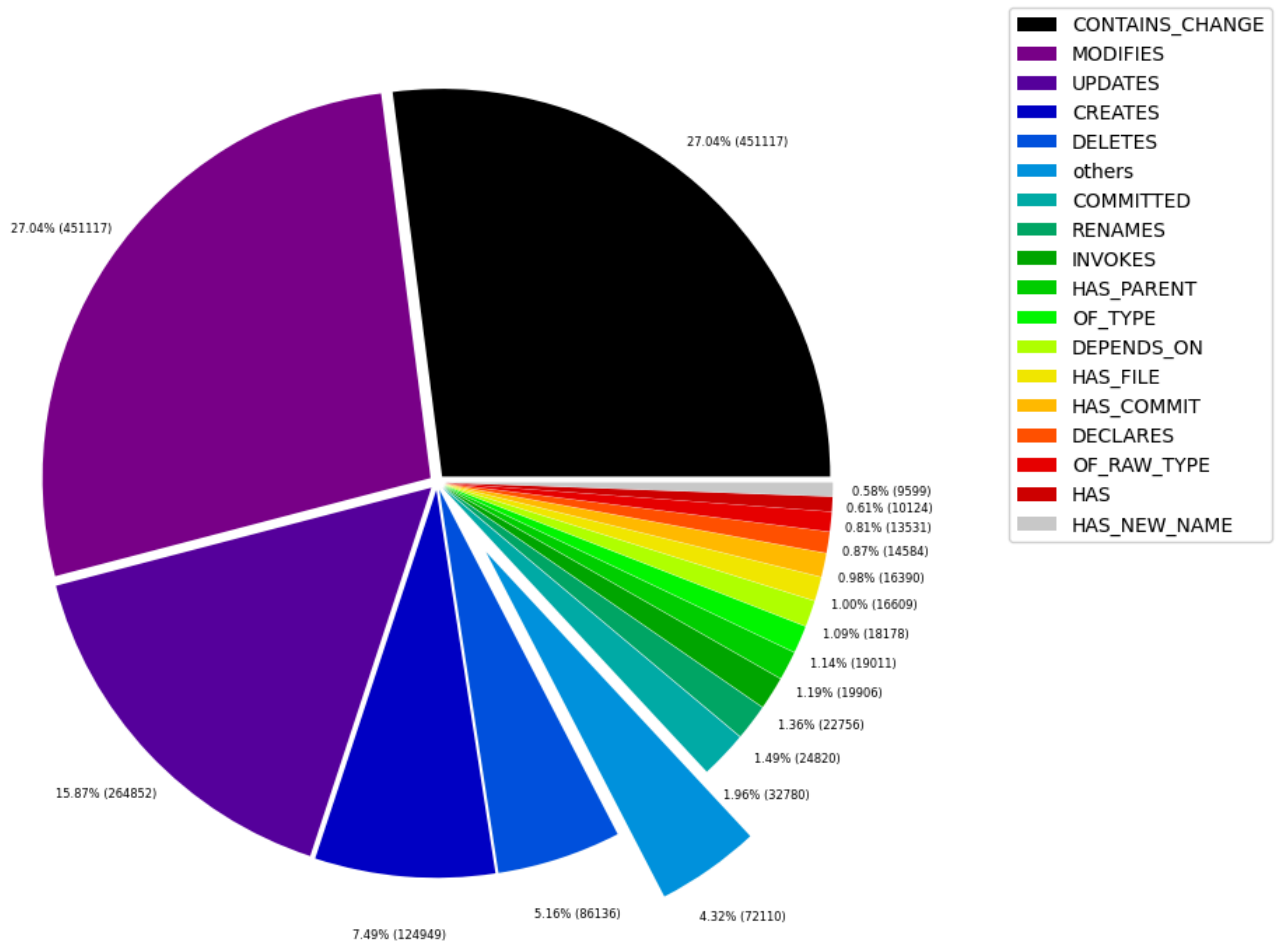


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

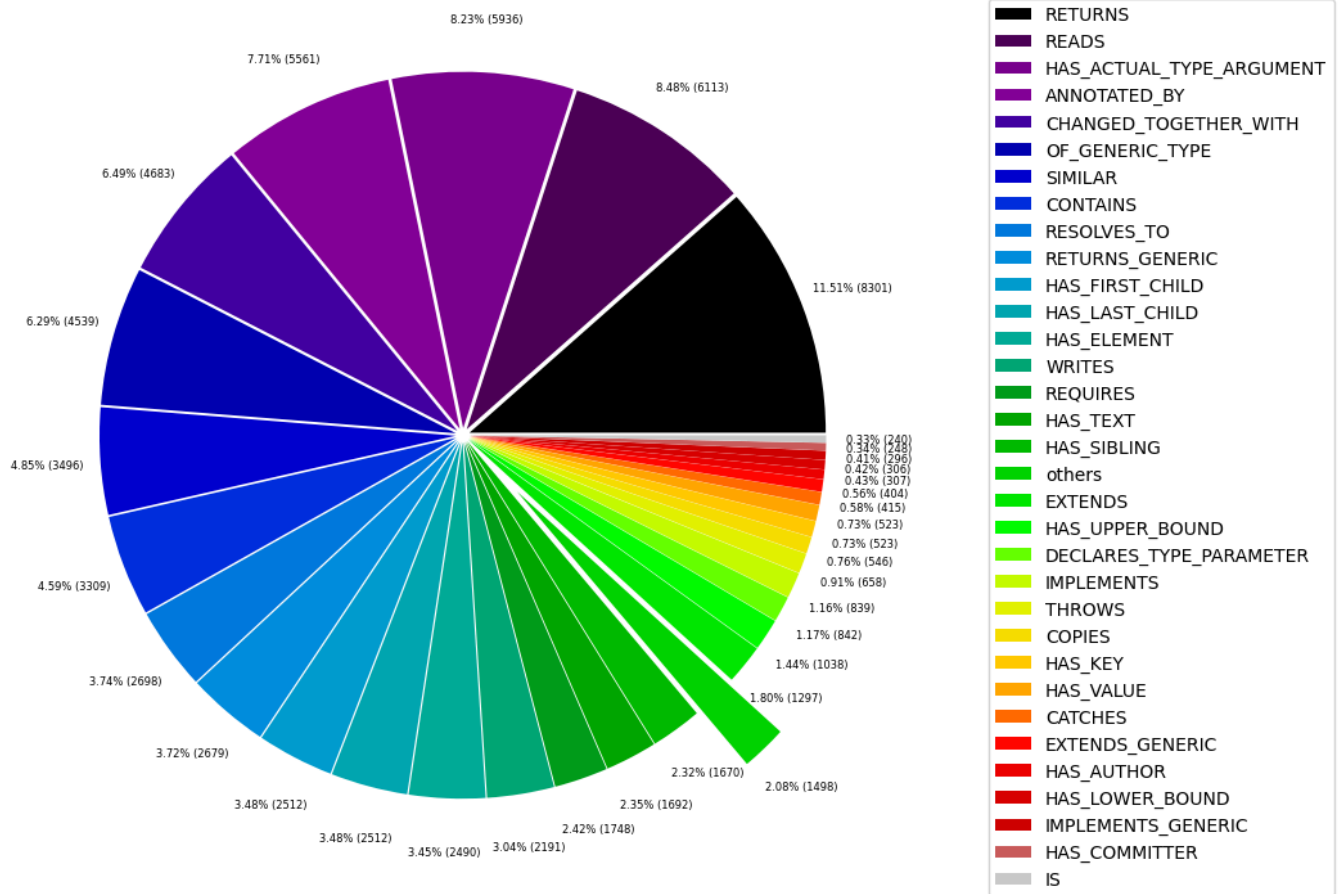
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000060
1	HAS_BRANCH	1	0.000060
2	HAS_HEAD	2	0.000120
3	THROWS_GENERIC	5	0.000300
4	EXCLUDES	6	0.000360
5	HAS_SCM	11	0.000659
6	DESCRIBES	11	0.000659
7	HAS_GOAL	13	0.000779
8	HAS_EXECUTION	13	0.000779
9	HAS_ROOT_ELEMENT	15	0.000899
10	INCLUDES_CONCEPT	19	0.001139
11	USES_PLUGIN	20	0.001199
12	HAS_CONFIGURATION	20	0.001199
13	IS_ARTIFACT	20	0.001199
14	HAS_ATTRIBUTE	22	0.001318
15	OF_NAMESPACE	22	0.001318
16	REQUIRES_TYPE_PARAMETER	26	0.001558
17	REQUIRES_CONCEPT	28	0.001678
18	DECLARES_NAMESPACE	44	0.002637
19	HAS_DEFAULT	54	0.003236
20	HAS_COMPONENT_TYPE	109	0.006533
21	CONTAINS_VALUE	126	0.007551
22	HAS_TAG	147	0.008810
23	ON_COMMIT	147	0.008810
24	DECLARES_DEPENDENCY	205	0.012286
25	TO_ARTIFACT	205	0.012286
26	COPY_OF	206	0.012346
27	IS	240	0.014384
28	HAS_COMMITTER	248	0.014863
29	IMPLEMENTS_GENERIC	296	0.017740

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberO
0	[Git, Update, Change]	UPDATES	[File, Git]	264852		264852
1	[Git, Update, Change]	MODIFIES	[File, Git]	264852		264852
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	264852		16390
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	99583		16390
4	[Git, Change, Create]	CREATES	[File, Git]	99583		99583
5	[Git, Change, Create]	MODIFIES	[File, Git]	99583		99583
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	61316		16390
7	[Git, Change, Delete]	MODIFIES	[File, Git]	61316		61316
8	[Git, Change, Delete]	DELETES	[File, Git]	61316		61316
9	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	24820		16390
10	[Git, Change, Rename]	DELETES	[File, Git]	24820		24820
11	[Git, Change, Rename]	MODIFIES	[File, Git]	24820		24820
12	[Git, Change, Rename]	CREATES	[File, Git]	24820		24820
13	[Git, Change, Rename]	RENAMES	[File, Git]	24820		24820
14	[Git, Commit]	HAS_PARENT	[Git, Commit]	19895		16390
15	[Repository, File, Git]	HAS_FILE	[File, Git]	16609		1
16	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	16390		1
17	[Author, Git, Person]	COMMITTED	[Git, Commit]	16390		307
18	[Committer, Git, Person]	COMMITTED	[Git, Commit]	16390		248
19	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	13642		9043
20	[File, Git]	HAS_NEW_NAME	[File, Git]	9599		16609
21	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	5528		9043
22	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	5443		9043
23	[Java, Value, ByteCode, Annotation]	OF_TYPE	[Type, File, Java, ByteCode, ExternalAnnotatio...	5201		5562
24	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	4218		9457
25	[Java, ByteCode, Parameter]	ANNOTATED_BY	[Java, Value, ByteCode, Annotation]	3915		9457
26	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	3885		16609
27	[Java, ByteCode, Bound, ParameterizedType]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	2884		5029
28	[Xml, Element]	HAS_ELEMENT	[Xml, Element]	2490		2512
29	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	2376		6545

Graph Density

```
total_number_of_nodes (vertices): 537822
total_number_of_relationships (edges): 1668569
-> total directed graph density: 5.7685658074180974e-06
-> total directed graph density in percent: 0.0005768565807418097
```